UNIZA 1 – Nature- Inspired Microstructures

Martin Mathew

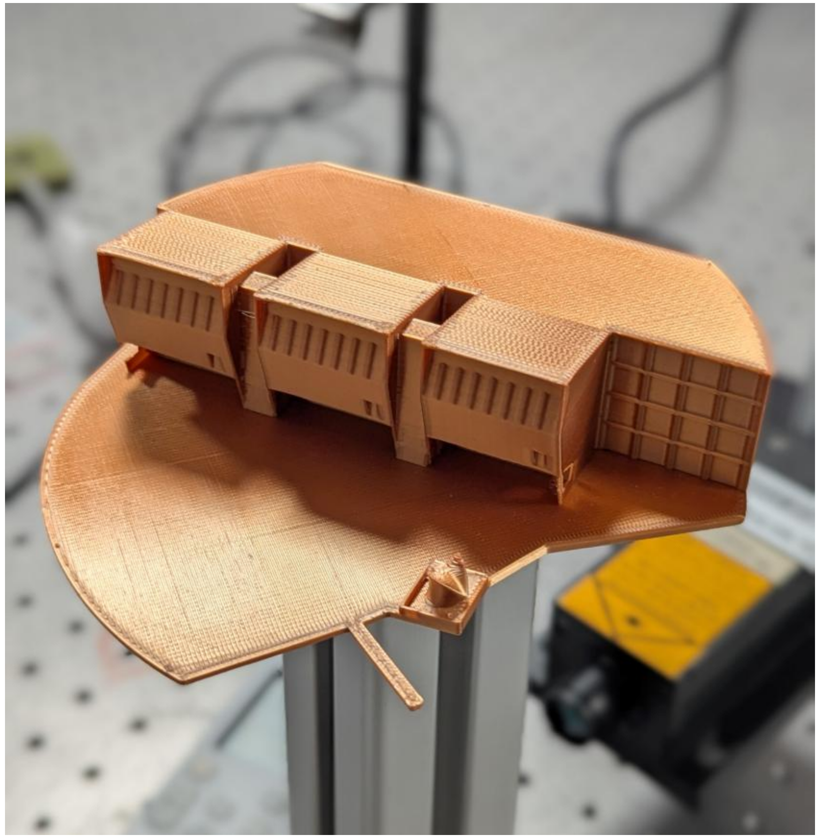
Dušan Pudiš , Peter Gašo, Ivana Lettrichova

TECHNISCHE UNIVERSITÄT DRESDEN
UNIVERSITY OF ŽILINA
Wrocław University of Science and Technology

Co-funded by the European Union

Outline

1. Introduction
2. Project plan
3. Results
4. Summary & Conclusions
5. My message ☺



Introduction

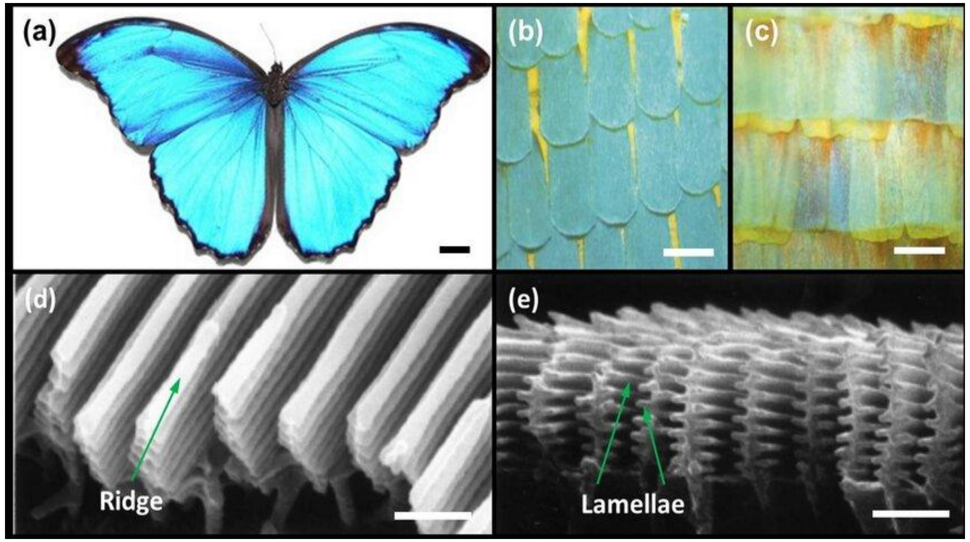
Introduction to the topic

Goal:

- Design and simulate a Morpho butterfly–inspired microstructure that reproduces its unique optical effects - primarily **structural colour and directional light scattering** — using a **parametric 3D model**

- Concept is inspired by the **nanostructure of Morpho butterfly wings**, which are well-known for producing **brilliant iridescent blue coloration** purely from **micro- and nano-scale structures**, not pigments

- **Fabrication limitations**: achieving microscale precision of lamella thickness (≈0.4 µm) and gap uniformity with 3D printing can be challenging



Source

Why did I choose it?

- Inspired by **nature's ability to create complex optical effects**
- Topic combines **bioinspired design, optical materials,** and **computational modelling**
- Study how structural geometry controls light - color coatings, sensors, and photonic devices
- Transition from **digital design** to **experimental fabrication** and **optical testing**

Co-funded by the European Union

Project plan

Project Plan

1. Research about Morpho
2. Plan modelling technique

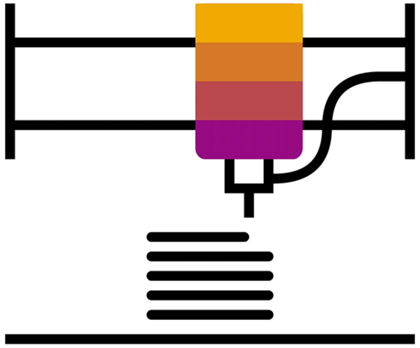
Primary Phase

1. Design Morpho-Inspired geometry using CAD
2. Export STL

Secondary Phase

1. Fabrication
2. Testing

Hands- On / Final Phase

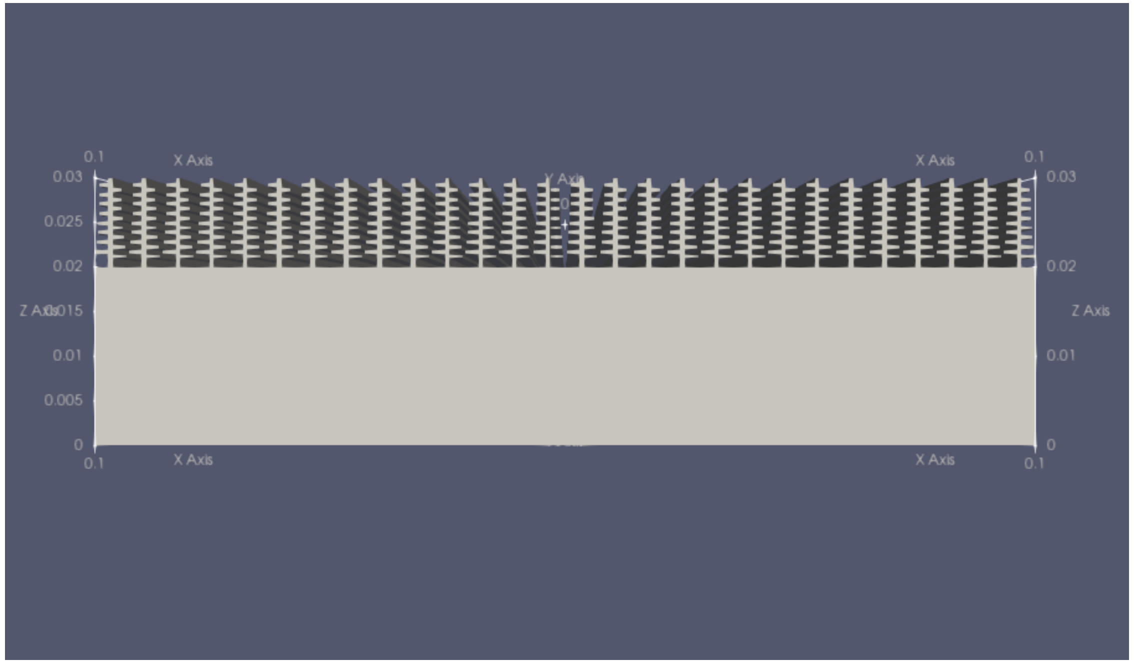


Project Plan

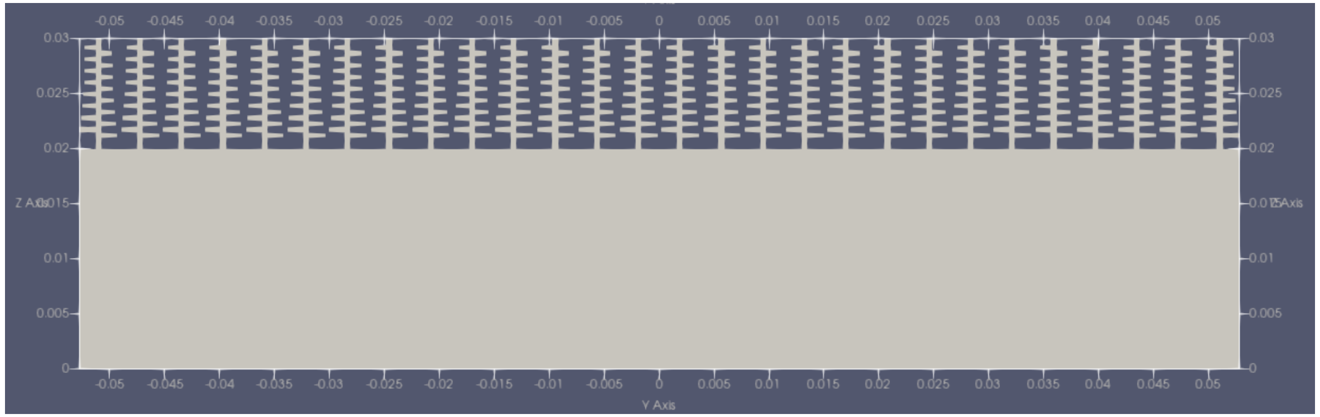
STEP 1 – Digital Design & Morphology Modeling
Objective: Create the biomimetic structure inspired by Morpho butterfly scales

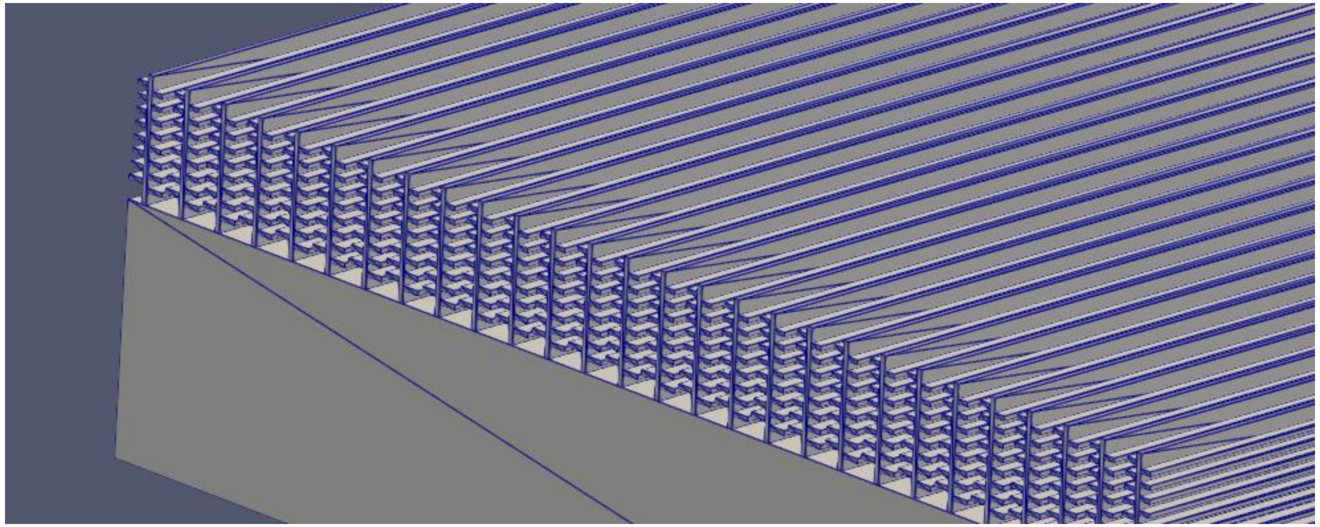
morpho.py
Python > morpho.py > ...

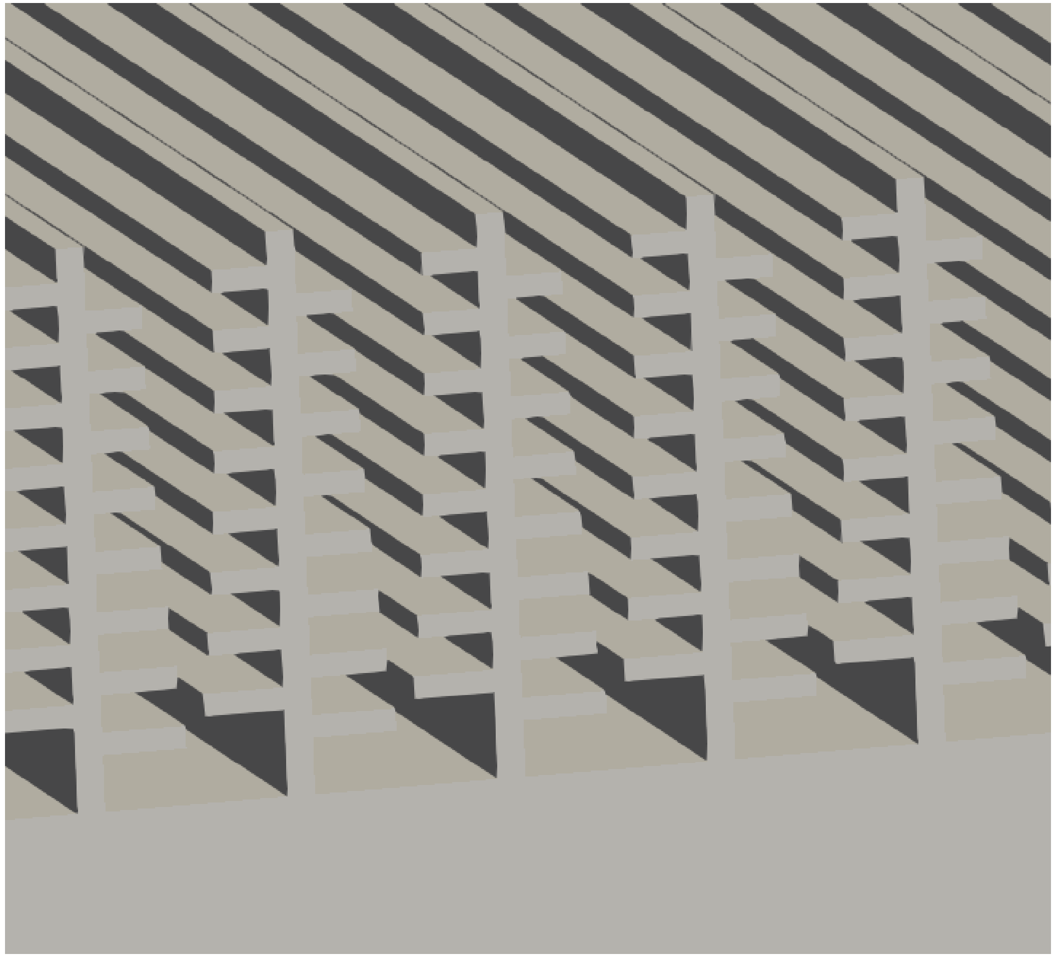
```python
"""

import trimesh
import os
import numpy as np
import pyvista as pv

# ---------------- Parameters (µm) ----------------
um_to_mm = 1e-3

# User parameters
n_trees_x = 1
n_trees_y = 28

tree_extrusion_um = 100.0
substrate_thickness_um = 20.0

# Backbone / lamella geometry
backbone_thickness_um = 0.5
backbone_height_um = 10.0
extrusion_length_um = tree_extrusion_um
lamella_thickness_um = 0.4
lamella_gap_right_um = 0.1
```



Project Plan





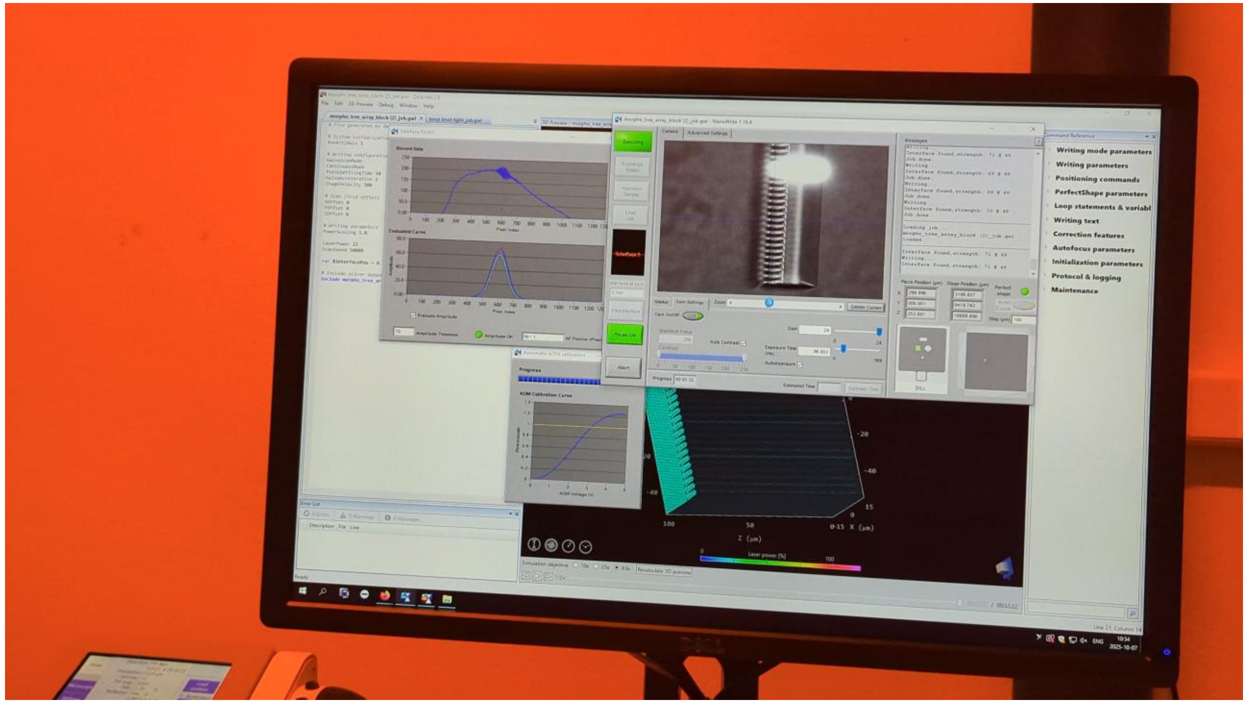


Project Plan

STEP 2 – Material Selection & 3D Printing
Objective: Fabricate the designed structure with high resolution

Tools & Methods:

- **Material:** IP- Dip
- **Technique:** Two-photon polymerization (TPP) 3D printing
- **Printer:** Nanoscribe
- **Post-processing:** Develop in alcohol solution



Project Plan

STEP 3 – Optical Measurement & Characterization
Objective: Evaluate the structural color and light interaction experimentally

Tools & Methods:

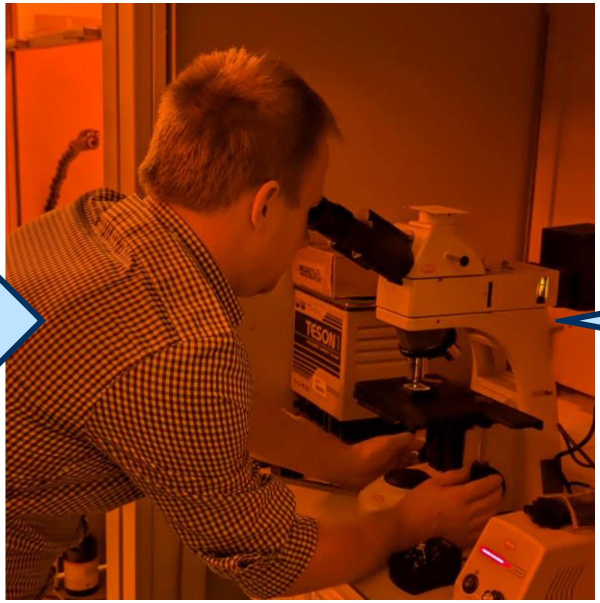
- **Measurement Set-up:** Optical bench with white light source, spectrometer, and adjustable incident angle
- **Device:** Ocean Optics spectrometer
- **Procedure:** Illuminate sample at normal and oblique angles; record reflected spectra in 400–700 nm range
- **Analysis:** Compare experimental reflectance spectra with simulated results
- **Software for Analysis:** Python (Matplotlib, NumPy) for data visualization and comparison

Results

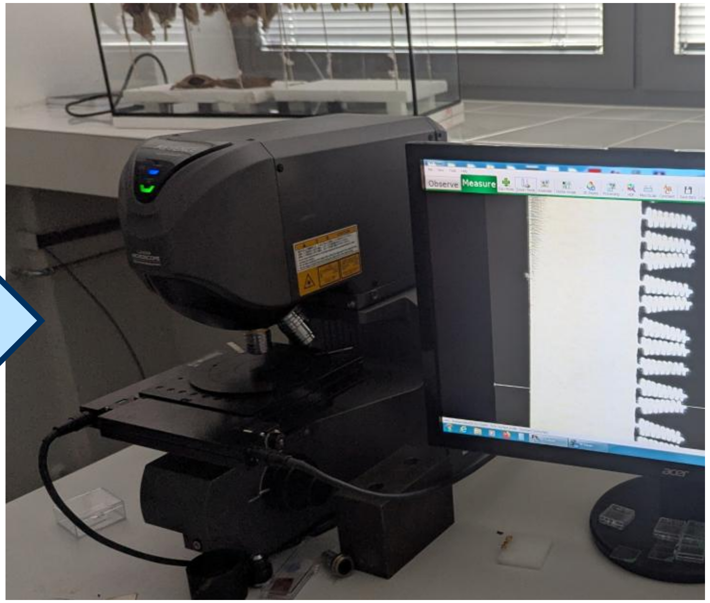
Co-funded by the European Union

Results/ Post- Processing

Developing Stage



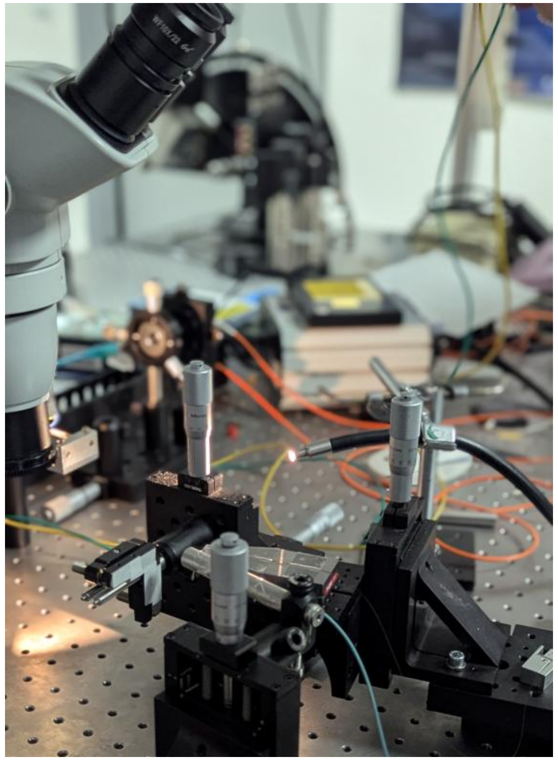
Preliminary Inspection using the Confocal Microscope



Inspection using the Laser Microscope

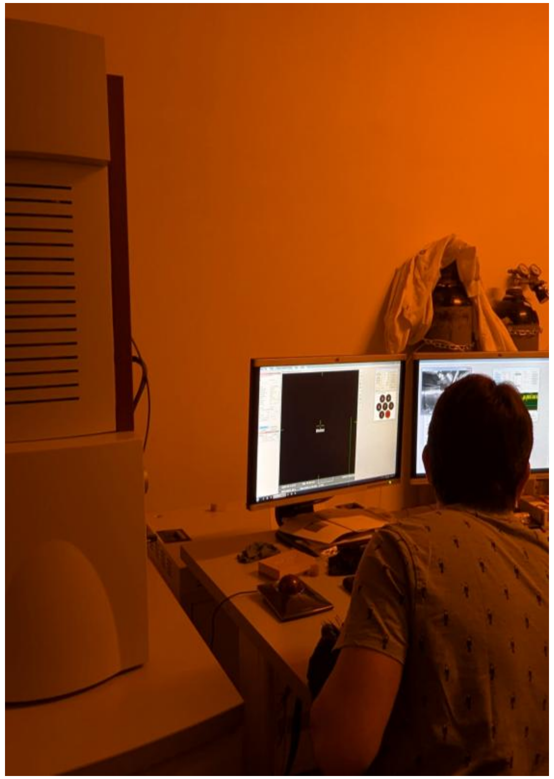
Co-funded by the European Union

Results/ Post- Processing



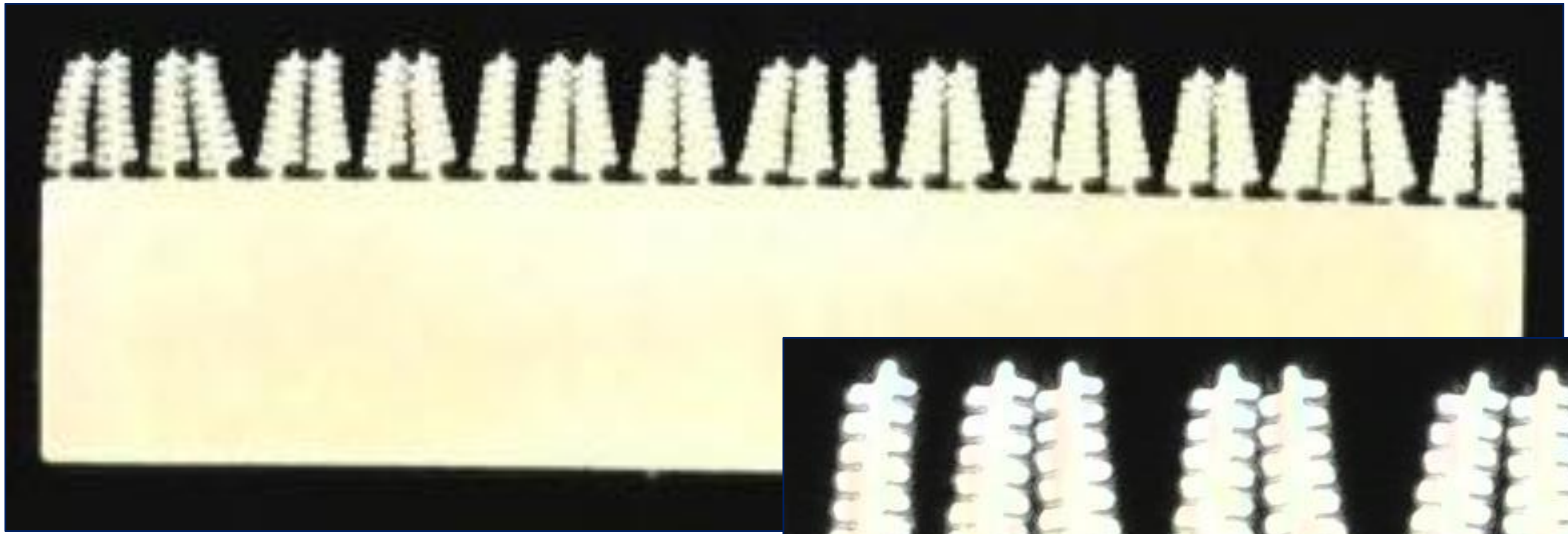
Spectroscopy Setup

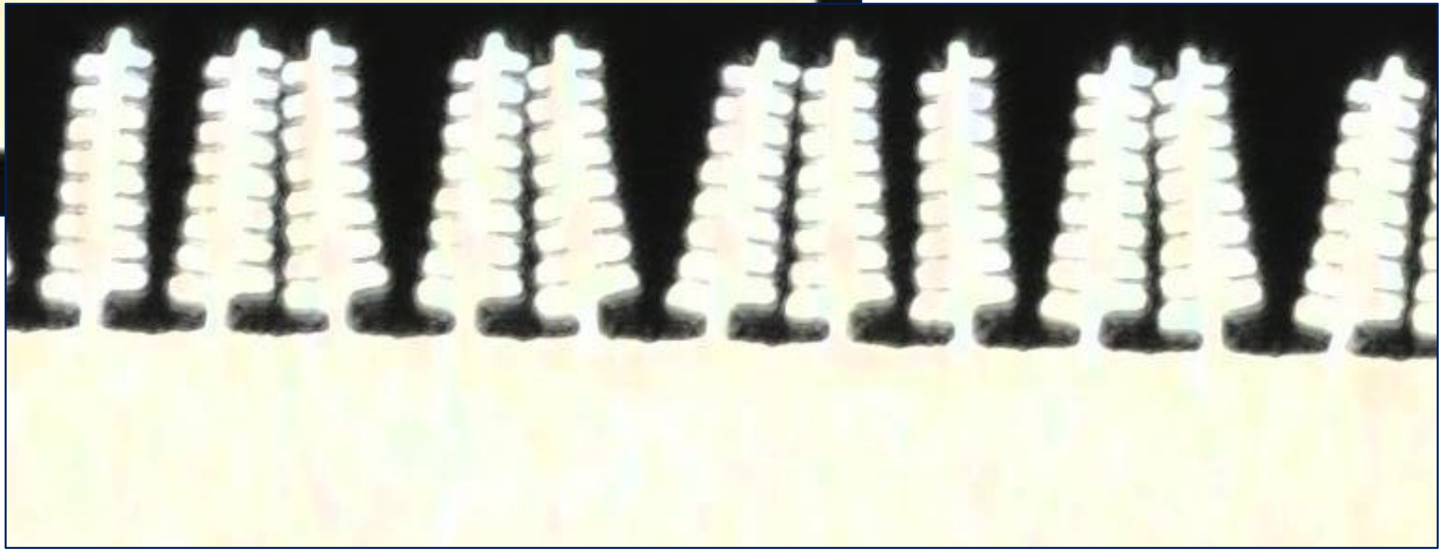
Spectral Analysis



Inspection using the Electron Microscope

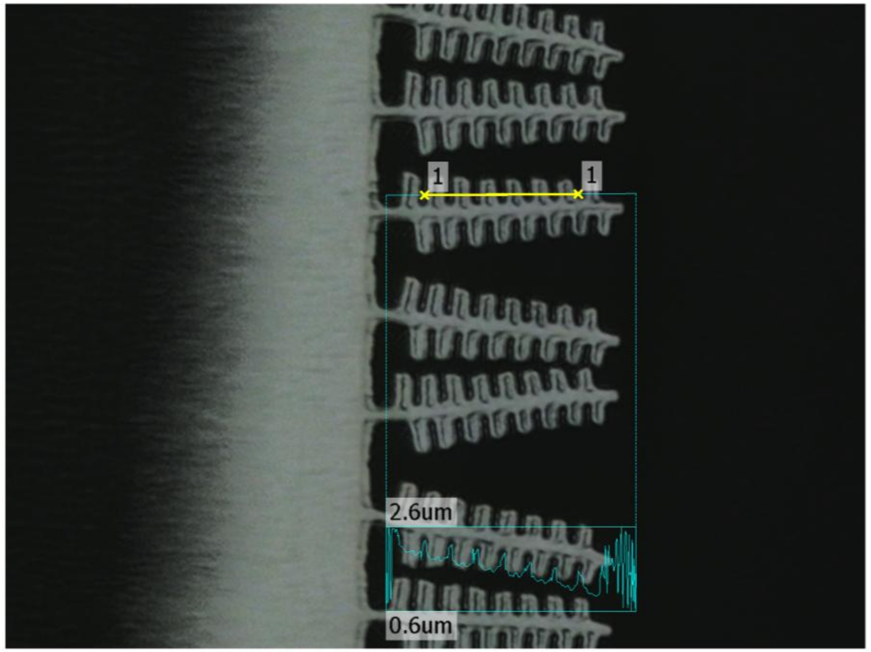
Results/ Post- Processing

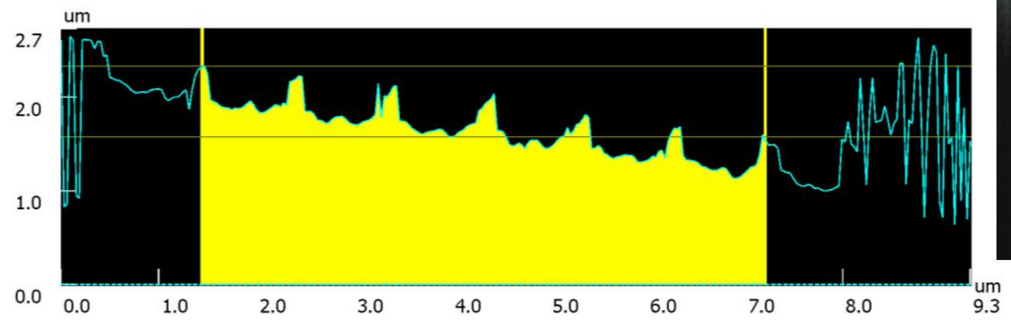


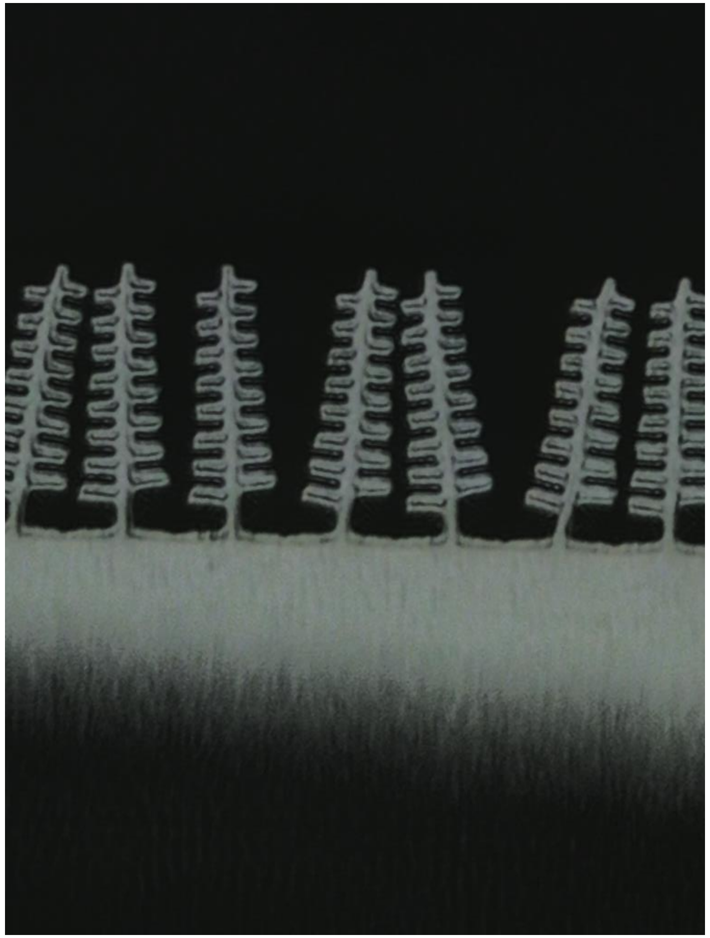


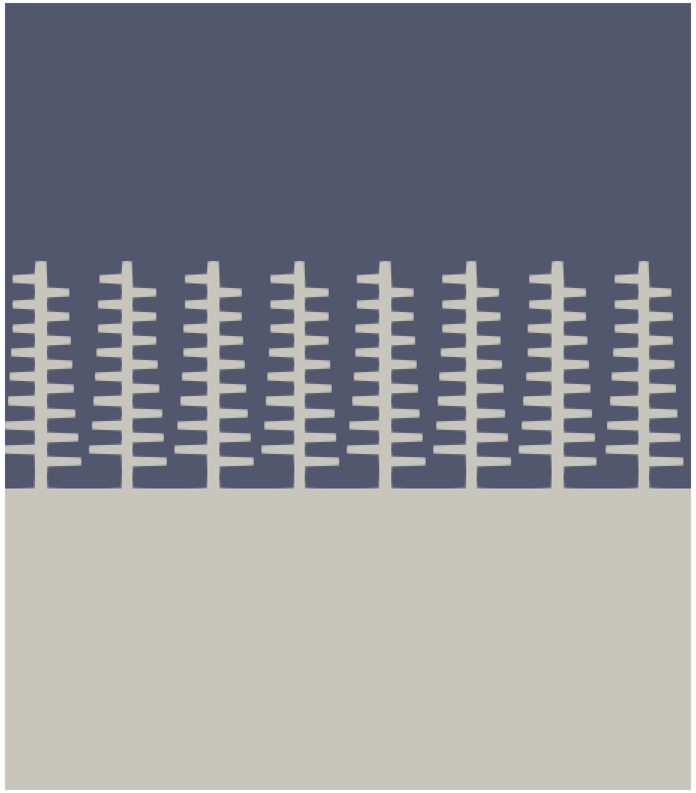
Images from Laser Microscope

Results/ Post- Processing





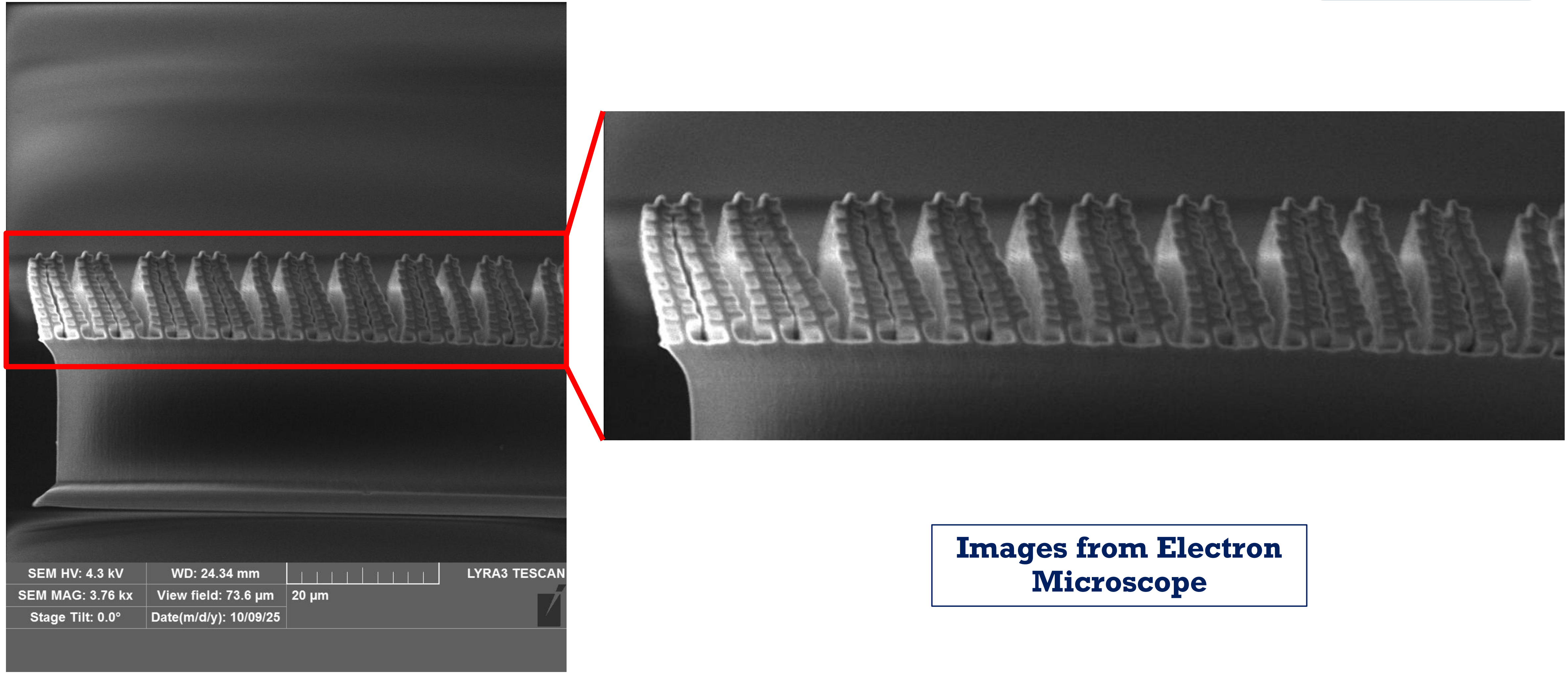




Period Length ~ 960 nm

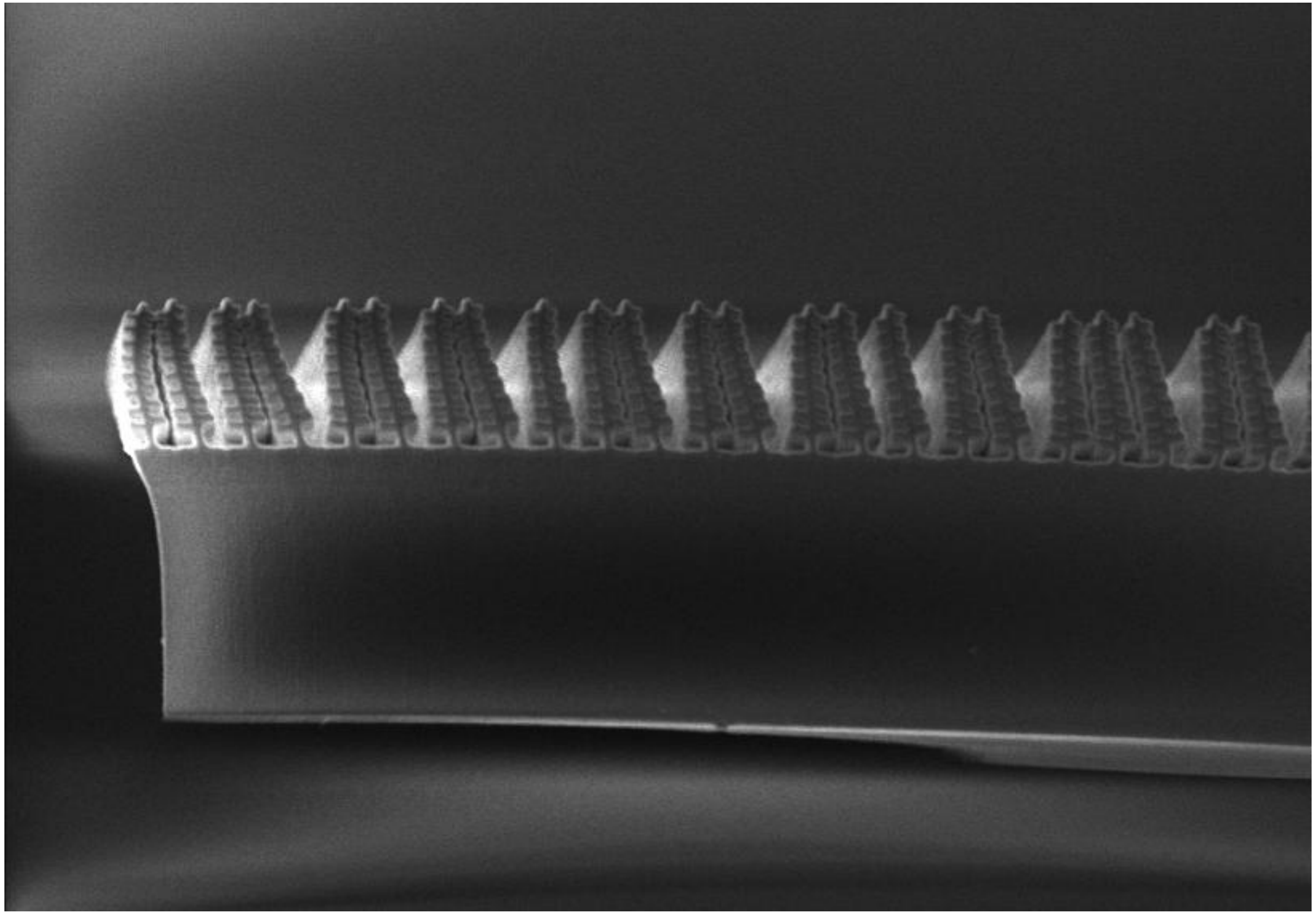
Co-funded by the European Union

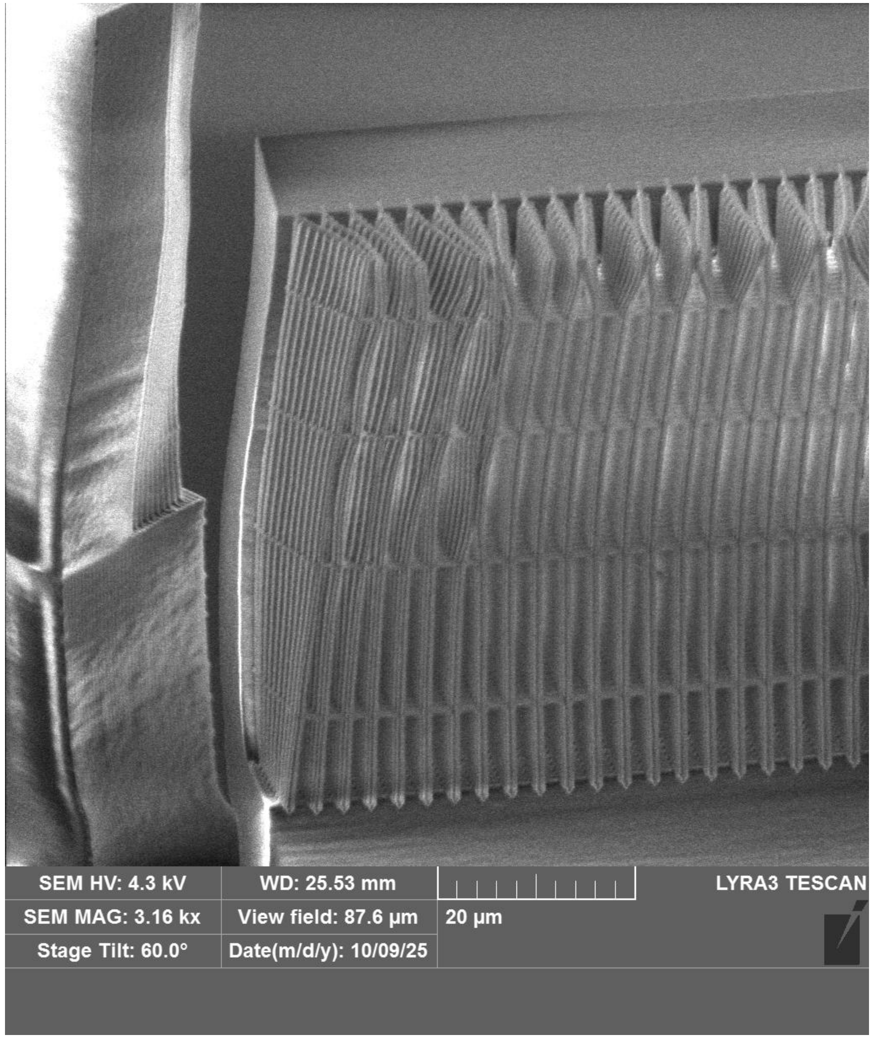
Results/ Post- Processing



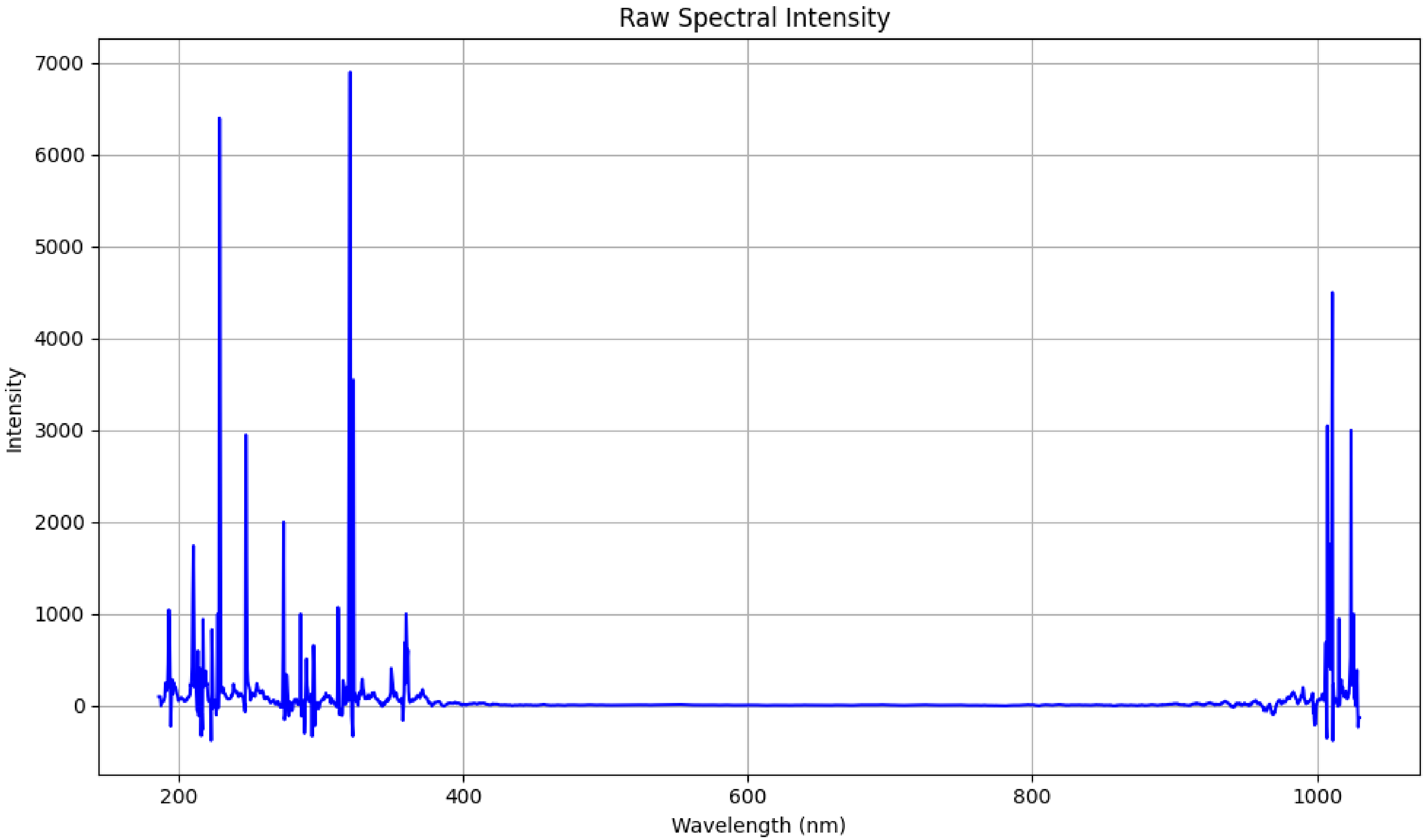
Images from Electron Microscope

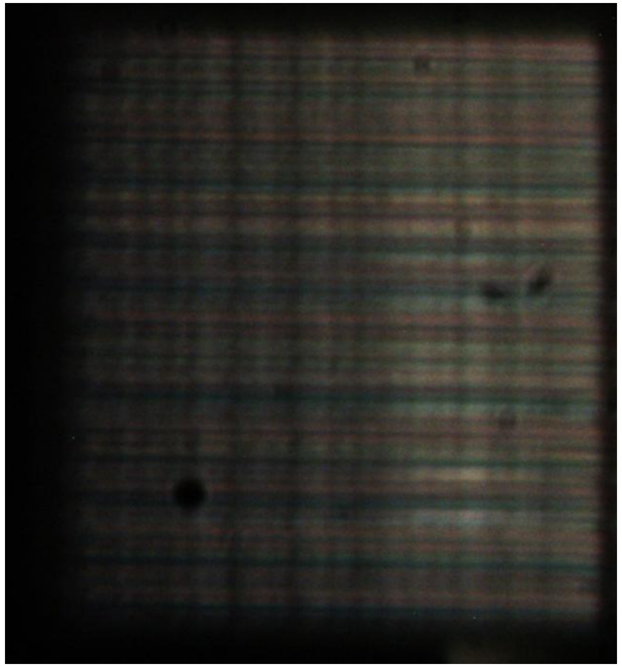
Results/ Post- Processing





Results/ Post- Processing





Summary & Conclusion

Summary & Conclusion

- The goal of the project was to **design** and **analyze** a **3D Morpho-inspired** structure that produces structural coloration through light interference
- Achieved by combining **Python**-based digital modelling
- Printed samples were prepared to test how surface morphology affects reflected color
- Optical measurements were performed using **Ocean Optics SpectraSuite** to record reflectance spectra (400–700 nm)

- The main project goal was **successfully** achieved
- **Ridge spacing** and **lamella height** are critical parameters controlling color shift and intensity

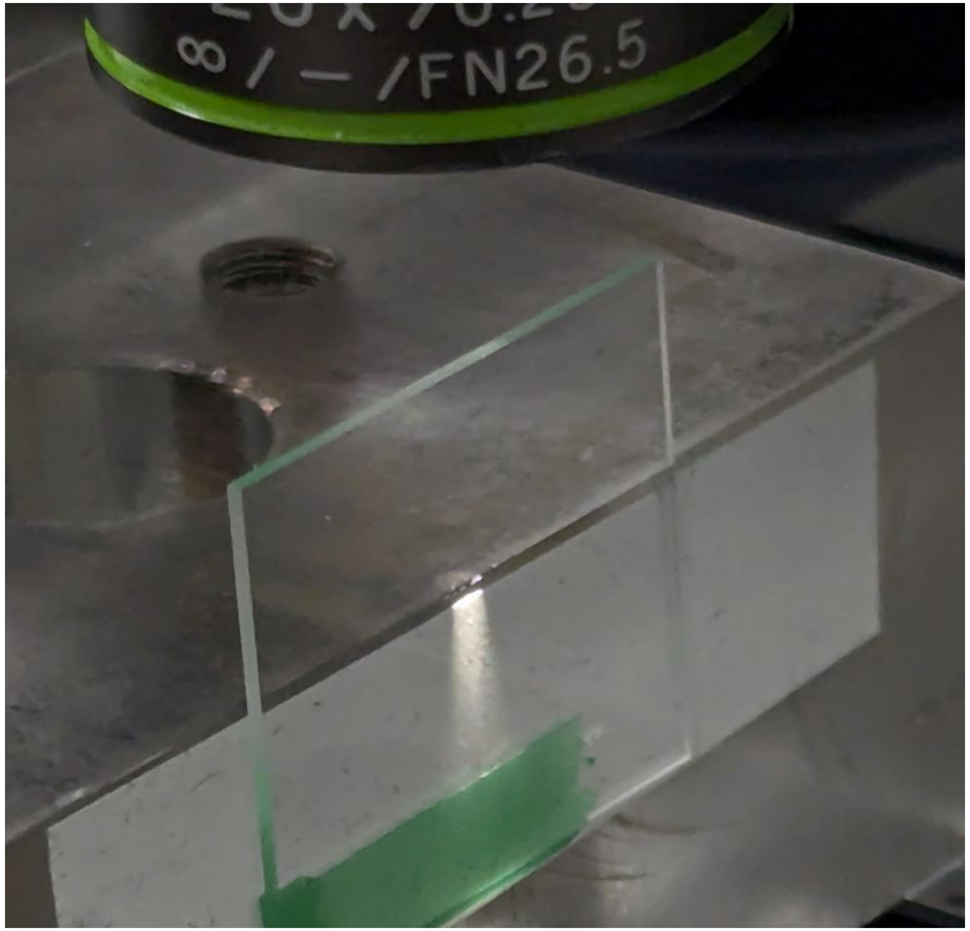
My message ☺

Co-funded by the European Union

Final Remark

- **Hands-on** experience bridging **digital modelling**, **additive manufacturing**, and **experimental optics**

- Strengthened my understanding of how **biological principles** can inspire **engineered optical surfaces**



Thank You!